

BA820 – Project M2

- **Section A1, Team 7**
 - **Student Name:** Keane Albright
- **Project Title:** Uncovering Structural Patterns in High-Stakes Regulatory and Funding Decisions for Drugs (European Drug Development)

1. Refined Problem Statement & Focus (~0.5 page)

The domain question investigated in this milestone is:

Can medicines be grouped by patterns of missing regulatory data that reflect different approval eras or regulatory contexts?

This question is reused verbatim from the M1 proposal. Its motivation remains unchanged: preliminary EDA suggested that missing regulatory information in the EMA dataset is structured rather than random, potentially reflecting differences in regulatory practices or documentation standards over time.

While the problem statement itself did not change, its operational framing became more concrete in this milestone. Rather than treating missingness descriptively, missing data patterns were explicitly modeled as features for unsupervised analysis. This shift enabled medicines to be compared based on shared data availability profiles rather than observed regulatory outcomes or therapeutic attributes.

Several assumptions from the proposal were evaluated. The assumption that missingness is non-random was validated, as consistent sparsity patterns emerged across medicine groups. The assumption that these patterns relate to approval timing or regulatory context was partially supported, as missingness-based groupings differed in approval year distributions. However, assumptions about specific causal drivers remain untested and were not inferred at this stage.

Finally, while the broader project includes multiple questions and datasets, this milestone narrowed its scope to Q2 and the drugs dataset. This refinement was driven by the strength and clarity of the missingness signal observed during early analysis.

2. EDA & Preprocessing: Updates (~0.75 page)

In M1, exploratory analysis revealed uneven and patterned missingness across regulatory variables in the drugs dataset. Certain fields were consistently absent for large subsets of medicines, with patterns suggesting differences across approval periods rather than random omission. These findings directly motivated a deeper exploration of missingness structure in this milestone.

Additional EDA focused on visualizing and summarizing missingness at the drug level. A missingness matrix confirmed that sparsity patterns co-occur across variables, reinforcing the idea that missingness reflects shared regulatory or temporal characteristics. Summary plots further showed that some regulatory fields were nearly complete for more recent approvals but frequently absent for older ones.

To support unsupervised analysis, each medicine was encoded as a binary missingness vector, indicating the presence or absence of each regulatory field. This transformation was necessary to treat missingness as the primary analytical signal. No normalization or scaling was applied, as all features were binary and on the same scale. This choice was intentional, with normalization deferred to later milestones when heterogeneous feature types are introduced.

No observations were removed due to missingness, as doing so would conflict with the analytical goal of studying missingness itself.

3. Analysis & Experiments (~1.5 page)

To assess whether medicines can be grouped by shared missingness patterns, K-Means clustering was applied to the binary missingness vectors. This method was selected as a baseline unsupervised approach to test whether structured grouping exists in the absence of labels.

K-Means is appropriate for this exploratory stage because it provides a clear and interpretable way to identify potential cluster structure. Although it assumes spherical clusters and relies on Euclidean distance, it serves as a reasonable starting point before considering more complex alternatives.

The number of clusters was explored using the elbow method, examining within-cluster variance across candidate values of K. Several values appeared plausible, and a small number of clusters was selected to balance interpretability with explanatory power. Cluster assignments were then merged back into the original dataset for interpretation.

To evaluate whether clusters reflected meaningful differences, approval year distributions were examined across clusters. Rather than using approval year as an input feature, it was treated as an external reference variable. This revealed that clusters differed systematically in their temporal composition: some were dominated by older approvals with higher overall missingness, while others consisted largely of more recent approvals with more complete regulatory records. Heatmaps of average missingness by cluster further highlighted distinct sparsity profiles across regulatory fields.

However, several limitations emerged. Cluster interpretation beyond high-level temporal differences was challenging, and clusters could not be cleanly mapped to specific regulatory pathways or policy regimes without additional features. The choice of K remained somewhat subjective, and different values produced slightly different but still plausible groupings. These results suggest that while clustering was informative, it is not sufficient on its own to explain the observed structure fully.

A notable insight from this analysis was how strongly missingness alone aligned with approval timing, even without explicitly incorporating temporal variables. This reinforced the idea that regulatory data collection practices evolve in detectable ways.

4. Findings & Interpretations (~0.75 page)

This milestone demonstrates that missing regulatory data in the EMA dataset is structured and informative. Medicines can be grouped based solely on patterns of data availability, and these groupings differ meaningfully in their approval year distributions. This suggests that missingness reflects broader regulatory or documentation contexts rather than random data loss.

From a practical perspective, these findings highlight the importance of treating missingness as a potential signal. For regulators and analysts, ignoring or automatically imputing missing values may obscure meaningful historical or procedural differences embedded in the data. More broadly, the results show that unsupervised methods can uncover latent structure even when working with absence rather than observed values.

These findings remain exploratory. While they support the hypothesis that missingness encodes regulatory context, they do not establish causal explanations. Nevertheless, they provide evidence that further unsupervised investigation is warranted.

5. Next Steps (~0.25 page)

Several elements remain unaddressed. Missingness-based clustering has not yet been compared to alternative methods such as hierarchical clustering, which may better capture non-spherical structure. Dimensionality reduction techniques could also be applied to support visualization and diagnostic assessment.

Future milestones will explore richer feature representations that combine missingness indicators with observed regulatory attributes, requiring normalization and additional preprocessing. Open questions include whether specific regulatory pathways align with missingness-based clusters and how these patterns evolve across datasets or domains.

The results of this milestone justify these next steps by demonstrating that missingness contains meaningful structure rather than being purely a data quality issue.

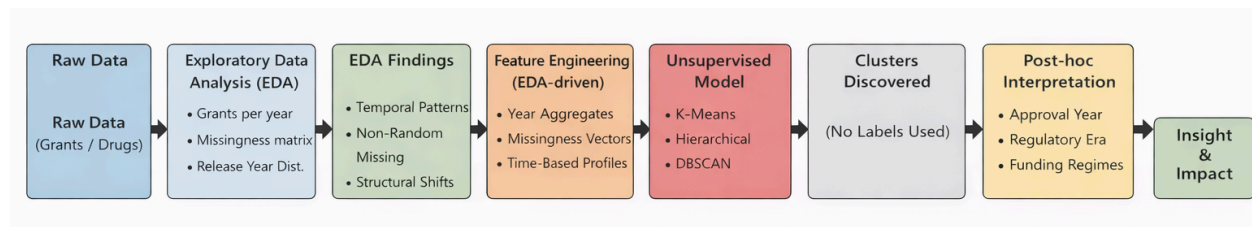
Appendix

GitHub Repository:

<https://github.com/EmperorRP/ba-820-team-a107.git>

My files are under the 'Keane' branch within the shared GitHub repository.

Process Overview:



A high-level process flow for an anticipated ML workflow. Subject to change depending on how the project and data patterns evolve.

External References and Context

For Drugs Dataset (Primary Dataset)

Regulatory pathways - <http://ncbi.nlm.nih.gov/books/NBK609384/>

Marketing Authorization by the EMA

<https://www.ema.europa.eu/en/human-regulatory-overview/marketing-authorisation>

Supplemental Material (Highly Recommended)

Use of Generative AI Tools

Generative AI tools (ChatGPT and Gemini) were used as assistive aids for the M2 deliverable. Their use was limited to clarification and drafting support, and did not replace my independent analysis, interpretation, or decision-making.

Specifically, AI tools were used in the following ways:

- Writing refinement: To assist with restructuring and tightening written sections for clarity and coherence, after the team had already determined the analytical direction, questions, and interpretations.

- Technical support: To help debug coding errors and clarify error messages encountered during exploratory analysis in Python.
- Insight understanding: Used to help understand/uncover insights from the notebook

All AI-generated suggestions were critically evaluated, edited, and integrated by me. Final dataset selection, EDA conclusions, domain questions, and interpretations were determined by me based on direct interaction with the data and the use of course concepts.

Documentation of AI interactions used during the process is provided below for transparency:

- ChatGPT conversation: <https://chatgpt.com/share/697983c6-7c8c-8009-9225-9ff45fee1f1f>
- Gemini conversation: <https://gemini.google.com/share/ceb54f8af9fe>